Workflow and architecture patterns for working together 2: Going forward

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June 7, 2026

Outline

1. Current challenges and their canonical solutions

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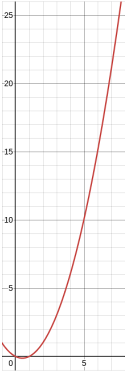
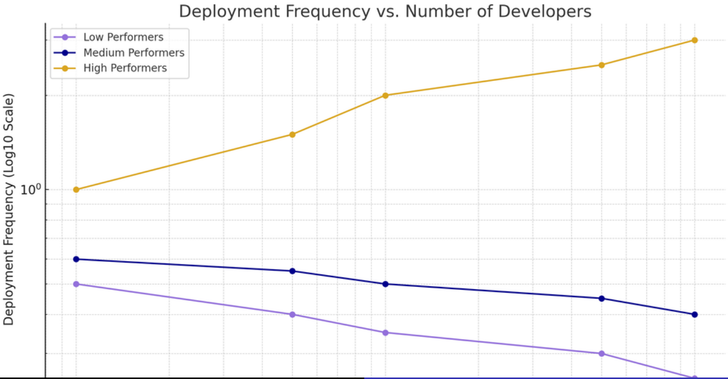
1975: The Mythical Man Month, Fred Brooks

- number of direct communication paths for n individuals$=$ $n(n-1)/2$



The Mythical Man Month, Fred Brooks (cont.)

- corollary: adding more people to a project can lead not only to diminishing returns on delivery speed, but to objectively less work being completed



Challenges

- multiple teams working in the same repository
- lack of direct collaboration, and occasionally trust, between development, operations, and testing
- late integration of contributor code
  - feedback delays
  - merge conflict regressions
  - sunk costs fallacy risks
- coupling of all services at the database layer
  - high storage costs
  - single point of failure risk
  - risk of cross-project interference

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- late integration $\rightarrow$ adopting a true continuous integration workflow
- database coupling $\rightarrow$ Bezos API mandate/"Microservices"

1967: Conway's law

'Organizations which design systems [...] are constrained to produce designs which are copies of the communication structures of these organizations.' - Melvin Conway, 'How do Committees Invent?' Datamation, 1967

Conway's law: examples

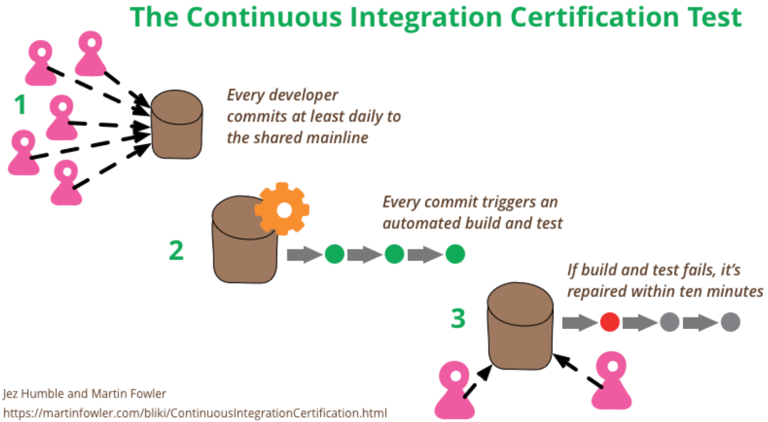
- separate `.csproj` projects for tests written by QA
- separate development and config repositories for the same transaction signifying low communication between development and operations
- all business documentation in the same repository, separate from the repositories that the documentation is for

Conway's law: corollaries

- Conway's law can't be fought
- Teams working in the same part of the same codebase at the same time are the same team
  - separate the codebases for improved velocity
  - join the teams for improved code sharing and communication (Inverse Conway maneuver)
  - branching-based strategies for team management compromise between the above approaches without their benefits

Outline

Defining continuous integration



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A naive probability theory for merge consistency: isolated commits

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A naive probability theory for merge consistency: modeling branches as sets of commits

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A naive probability theory for merge consistency: general definition for distinct branches diverging from a common ancestor

- Let A be the set of unmerged commits across k distinct branches $A_1 \cdots A_k$, where $A = A_1 \sqcup A_2 \sqcup \cdots \sqcup A_k$, $n = |A|$, and $n_i = |A_i|$ for $1 \leq i \leq k$. Then

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$$P(no\ conflicts) = \frac{1}{2^n - (\sum_{i=1}^{k} 2^{n_i}) + k}$$

The State of DevOps report: branching findings

We found that having branches or forks with very short lifetimes (less than a day) before being merged into trunk, and less than three active branches in total, are important aspects of continuous delivery, and all contribute to higher performance. So does merging code into trunk or master on a daily basis.
— The 2016 State of DevOps Report, p. 31

Outline

The State of DevOps Report: throughput metrics

- lead time for changes

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- lead time for changes
- deployment frequency

The State of DevOps Report: stability metrics

- change failure rate

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- mean time to restore

The State of DevOps Report: results

'Astonishingly, these results demonstrate that there is no trade-off between improving performance and achieving higher levels of stability and quality. Rather, high performers do better at all of these measures. This is precisely what the Agile and Lean movements predict, but much dogma in our industry still rests on the false assumption that moving faster means trading off against other performance goals, rather than enabling and reinforcing them.' - Nicole Forsgren, Jez Humble, and Gene Kim, Accelerate (2017), p. 14

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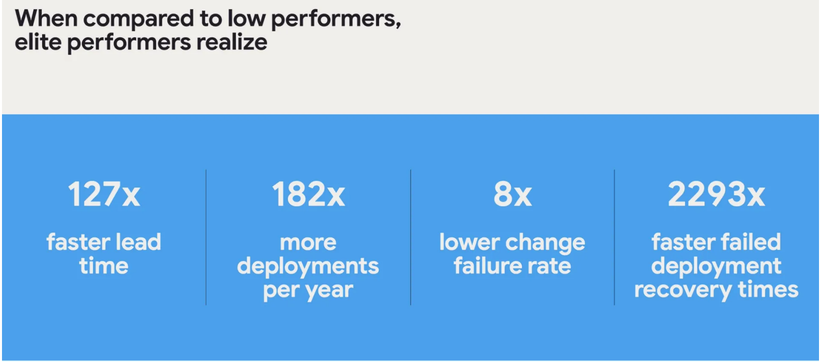

— 2024 State of DevOps Report

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PEDI-35120_837D_COB	85	15 Apr	5
Either COB branch	159	23 Mar	6
All other branches	123	23 Mar	12
	72	15 Apr	10
	266	21 Jan	

Source: GH Actions Simulation tests, 28 May 2026

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All other branches	23/88 (26%)
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- 50% less merge work
- 25% less backmerge work
- Source command: `git log --since="2026-04-15" --oneline <--merges> <--all> <^PEDI-35120_837D_COB> | <grep -iv ''merge pull request''> | wc -l`

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- Phase one focuses on a) greater integration between developers in the same repository, and b) greater integration of development and testing, with later phases targeted at improving dev/ops collaboration and intra-team CloudOps capabilities

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Going forward: defining a team

- A *team* is the minimal socio-technical unit that can deliver a production change with no additional coordination beyond mandated governance or custodial controls.
  - pull requests
  - testing
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Workflow and architecture patterns for working together 2: Going forward

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